TISA SELMA

Applied AI Researcher | Networked Multimedia | Multimodal & Trustworthy ML

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Research profile

Computer-science researcher developing data-driven inference systems for human experience in networked multimedia. Research spans HTTP adaptive streaming, HTTPS/QUIC traffic, video quality of experience, facial-expression and engagement signals, drift-aware/self-healing prediction, and reproducible ML evaluation. More than sixteen years of public-sector IT work add an operational lens: privacy, security, reliability and maintainability matter alongside predictive performance.

Research interests

- Networked multimedia, adaptive streaming and encrypted-traffic inference
- Computer vision, affective computing and multimodal learning
- Trustworthy ML: drift, calibration, robustness, privacy and uncertainty
- Distributed/federated learning, research software and data-intensive systems

Education

Doctoral Research, Computer Science · United Arab Emirates University, Al Ain, UAE · Aug 2019-Jun 2026
Dissertation: *Data-Driven QoE Inference for HTTP Adaptive Streaming*

Master of Engineering, Image Processing · Inje University, Gimhae, South Korea · 2013-2015

Bachelor of Applied Science, Information Technology / Informatics · Telkom University, Bandung, Indonesia · 2004-2009

Research experience

Graduate Research Assistant & Doctoral Researcher · United Arab Emirates University · 2019-2026

- Designed studies that connect playback, network, advertising, facial-expression and feedback signals to perceived streaming experience.
- Developed machine-learning pipelines for QoE inference over HTTP adaptive streaming, including HTTPS and QUIC settings.
- Investigated participant-aware evaluation, domain shift, performance monitoring and self-healing model behavior.
- Authored peer-reviewed work across IEEE Access, Technologies and IEEE conference proceedings; prepared code, figures, analysis and manuscripts.
- Worked across Python-based ML, deep learning, signal/image processing and data-intensive experimentation.

Professional experience

Senior Information Technology Officer · House of Representatives of the Republic of Indonesia · Dec 2009-present

- Contributed to public-sector software delivery, information systems, cybersecurity and operational reliability in a complex stakeholder environment.
- Translated organizational needs into technical roadmaps, implementation work and cross-functional coordination.
- Supported long-lived systems where auditability, continuity, confidentiality and service quality are essential.

Selected publications

1. **From Facial Expressions to Engagement: A Comprehensive Multimodal Approach to Infer Video Streaming User Experience.** IEEE Access 13, 213886-213909, 2025.
2. **Inference Analysis of Video Quality of Experience in Relation with Face Emotion, Video Advertisement, and ITU-T P.1203.** Technologies 12(5), 62, 2024.
3. **Video QoE Inference with Machine Learning.** IEEE International Wireless Communications and Mobile Computing Conference, 2021.
4. **Inferring Quality of Experience for Adaptive Video Streaming over HTTPS and QUIC.** IEEE International Wireless Communications and Mobile Computing Conference, 2020.
5. **Contrast-enhanced Bias-corrected Distance-regularized Level Set Method Applied to Hippocampus Segmentation.** Journal of Korea Multimedia Society, 2016.

Selected research prototypes

QoE Foresight Repro · participant-grouped QoE evaluation — deterministic Python pipeline with training-only preprocessing, zero subject overlap, constant-baseline comparison, worst-subject error and synthetic public fixtures.

CipherQoE · encrypted-flow metadata inference — demonstrates bitrate-class, startup-delay and stall estimation from aggregate flow features with an explicit zero-payload boundary and privacy limitations.

FedQoE Bench · distributed learning systems — non-IID federated simulation tracking mean/worst-client error, partial participation, quantized updates and communication bytes.

TrustFlow Lab · trustworthy security analytics — time-ordered SOC evaluation with threshold calibration, false-positive reporting, abstention framing and benign-behavior drift monitoring.

All four projects use seeded synthetic fixtures and are presented as reproducibility demonstrations, not production or real-data benchmarks.

Technical skills

Machine learning: Python, TensorFlow/Keras, scikit-learn, deep learning, computer vision, multimodal inference, experimental design · **Data & distributed systems:** Scala, Spark, Hadoop, Neo4j, MongoDB, AWS, Google Cloud · **Software & methods:** C/C++, JavaScript, Node.js, SQL/NoSQL, grouped/temporal validation, drift monitoring, reproducibility